Optimal Timing of Power Play in Mixed Doubles Curling

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Background: Mixed Doubles Curling and Power Play

What is Mixed Doubles Curling?

- A game between **two teams**, each with **two players**
- The game is played over **8 rounds (known as ends)**
- Teams slide stones on the ice toward a **target**
- Each team throws **5 stones per end**, alternating turns
- At the end of each round, the team with stones closer to the center **scores points**
- Each round starts with **two stones in play**
- Game strategy: How to balance **offense (scoring)** and **defense (blocking)**?



Figure 1. A mixed doubles match (USA) in the 2022 Winter Olympics. Source: `https://x.com/usacurl`

What is Power Play (PP)?

- A special option that can be used **only once per game**
- At the beginning of an end, the team with the *hammer* (last stone of the end) can use PP.
- PP changes the starting layout to the **side of the target**
- Creates more space → **higher scoring potential**
- Increases risk → **opponent may score multiple points**

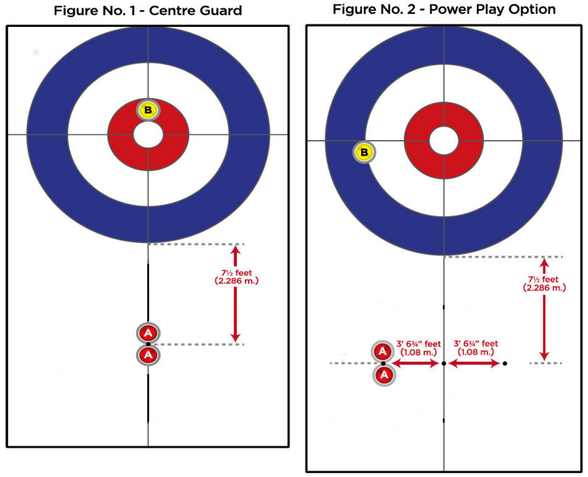


Figure 2. Standard vs. Power Play positioning. Source: `https://ardsleycurling.club/`

Question: When is the Best Time to Use Power Play?

Challenge: Using it early may help gain early lead, while saving it for later may be crucial when trailing.

Exactly when should we use this one-time option? It depends...

1. **Score difference**: far behind vs. slightly behind
2. **Remaining ends**: less time to fix mistakes with fewer ends left
3. **Who has the hammer?**: In each end, one team throws the final stone, known as the *hammer*. This is a key advantage, as the last shot often determines who scores points.
4. Team and opponent strength
5. Recent game flow (*momentum*)

Remark: For simplicity, we only considered factors 1-3.

Method: Finite-Horizon Stochastic Decision Model

Sequential decision under uncertainty: Before each end, teams reconsider whether to use PP if the option is still available.

Optimizing the win probability (value function): Suppose the game status after the tth end is $x = (\text{ends remaining}, \text{score_diff}, \text{hammer}, \text{PP}_{\text{you}}, \text{PP}_{\text{opp}})$, where PP_{you} and PP_{opp} indicate whether PP remains available to your or the opposing team. Then the win probability under optimal decision, denoted by $V(x)$, satisfies the **Bellman equation**,

$$V(x) = \max_a \mathbb{E}[V(X_{t+1}) \mid X_t = x, a], \tag{1}$$

where the expectation is with respect to the game transition $X_{t+1}|X_t$. The action a (i.e. using PP or not) maximizing the right-hand side of (1) is then the **optimal decision** under x. And the mapping $a = \pi^*(x)$ is called the **optimal policy**.

Computing the Optimal Policy by Dynamic Programming

Evaluating the expectation in (1) requires knowing the game transition $X_{t+1}|X_t$. Assuming it is known, the win probabilities $V(x)$ is solved by **dynamic programming**.

- **Base case** ($t = 8$, i.e. when the game ends): Given the game status x, $V(x)$ is directly known from the score difference.
- **Recursive step** ($t \to t-1$): For each state x, evaluate both actions (use PP or not), then find the maximum in Equation 1.

Remark: Note that, after $V(x)$ is computed for all possible game status x, the **optimal policy** $\pi^*(x)$ is simply a byproduct by following the actions that corresponds to the maximum at the right-hand side of (1).

Estimating the Game Transition from Past Tournaments

Data: The CSAS Data Challenge provides end-by-end game records of 344 mixed doubles matches from the 2022 Winter Olympics and 2023-2025 World Championships.

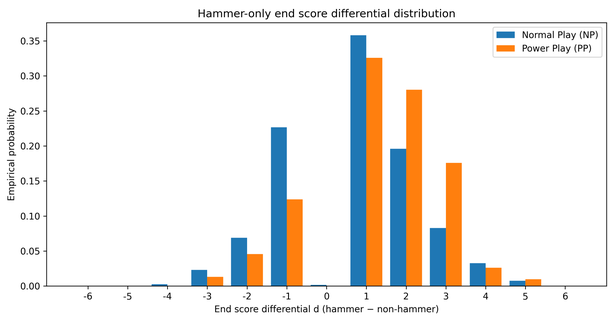


Figure 3. Score difference distribution.

- PP ends exhibit **more volatility** and in scoring compared to normal play ends

Empirical estimates of game transitions:

- Assuming the transition only depends on ends remaining and score differences, we used empirical counts in past tournaments to estimate the transition probabilities $P(X_{t+1}|X_t)$.

Possible extension: If other factors (e.g. team strength and in-game momentum) need to be considered, logistic regression or more advanced machine learning methods may be used to model the transitions.

How Reliable is the Optimal Decision?

Note that the estimation uncertainty in the transition probabilities brings uncertainty to the computed optimal policy.

- **Step 1**: We use a **Bayesian model (Dirichlet-multinomial model)** to derive the posterior distribution of the transition probabilities.
- **Step 2**: We draw 50,000 Monte Carlo samples from the Dirichlet posterior distributions in Step 1 and recompute the win probabilities $V(x)$ under each scenario, which gives us an approximation to the **posterior distribution of** $V(x)$.

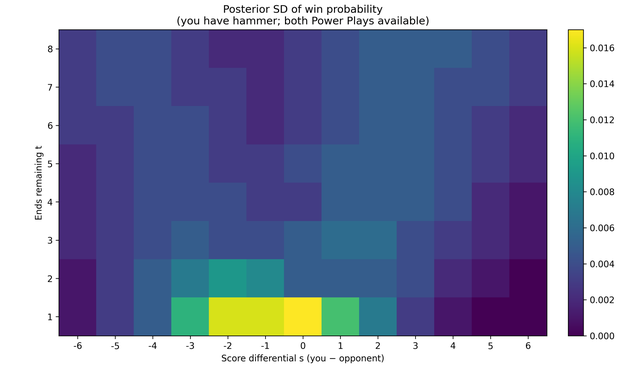


Figure 4. Estimated win probabilities for the hammer team when both teams have not used PP.

Remark: Figure 4 shows that the posterior standard deviation of the win probabilities **never exceeds 0.02**, indicating high confidence in our recommendations.

- **Step 3**: Based on posterior samples in Step 2, we can further obtain the **posterior probability of PP being the optimal action**.

Recommendation: Use Power Play When Trailing Earlier!

After thresholding the posterior probability of PP being the optimal action by 0.5, we obtain our recommended Power Play policy (Figure 5).

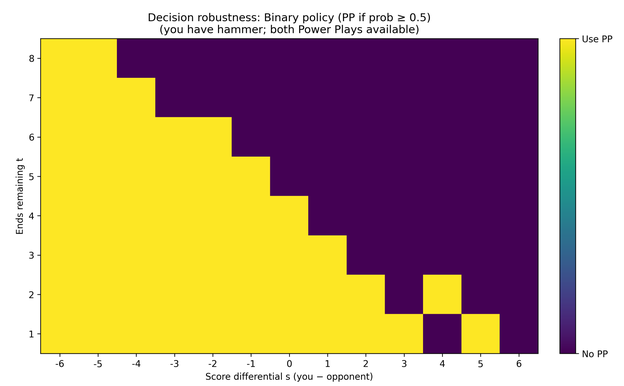


Figure 5. Optimal policy for the hammer team when PP is available for both teams (Yellow - use PP; Purple - do not use).

After comparing with the PP use in the tournament data, we found

- there is a good degree of agreement between our optimal policy and the actual PP use;
- however, teams should **use Power Play earlier** when trailing by 1 to 3 points with several ends remaining (e.g., $t = 4$, $s = -2$) according to our optimal policy.